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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (Minor in Robot Technology) Minor Degree Examinations**

May 2025

RI2009-1 - FUNDAMENTALS OF ROBOTICS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which factor is NOT generally considered when choosing a robot for a specific application?
A) Number of axes
B) Repeatability
C) Work volume
D) Color
2. Which of the following is an example of a fixed automation system?
A) Industrial robots
B) Transfer lines
C) Autonomous guided vehicles
D) Programmable logic controllers
3. In robotics, what is inverse kinematics used for?
A) Positioning the robot's base
B) Calculating motor power
C) Finding joint variables
D) Optimizing the robot's path
4. What is the workspace of a robot?
A) The area within which the robot can operate
B) The amount of memory available for programming
C) The size of the robot's end-effector
D) The set of instructions stored in the controller
5. What type of gripper is typically used to handle delicate objects, such as glass or soft materials?
A) Mechanical gripper
B) Vacuum gripper
C) Hydraulic gripper
D) Magnetic gripper
6. Given a point $a_{uvw} = (4, 3, 2)^T$ with respect to the rotated OUVW coordinate system, determine corresponding point a_{xyz} with respect to the reference coordinate system if it has been rotated 60° about OZ axis
A) $\begin{bmatrix} 0.598 \\ 4.964 \\ -2 \end{bmatrix}$
B) $\begin{bmatrix} -0.598 \\ 4.964 \\ 2 \end{bmatrix}$
C) $\begin{bmatrix} 4 \\ 1.26 \\ -2 \end{bmatrix}$
D) $\begin{bmatrix} -4 \\ -1.26 \\ -2 \end{bmatrix}$
7. A point $a_{uvw} = (2, 3, 4)$ is to be translated a distance -6 units along the OY axis and -4 units along the OZ axis. Using the appropriate homogeneous transformation matrix, determine the new point a_{xyz} .
A) $\begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$
B) $\begin{bmatrix} -2 \\ 3 \\ 0 \\ 1 \end{bmatrix}$
C) $\begin{bmatrix} -3 \\ -2 \\ -1 \\ 1 \end{bmatrix}$
D) $\begin{bmatrix} 3 \\ -2 \\ 1 \\ 1 \end{bmatrix}$

8. What is the inverse of a rotation matrix?
 - A) Its negative
 - B) Its reflection
 - C) Its determinant
 - D) Its transpose
9. Which of the following defines accuracy in a sensor?
 - A) The smallest measurable unit
 - B) The degree of closeness between the measured value and true value
 - C) The ability to return to the same value repeatedly
 - D) The range of the sensor's output
10. The triangulation method in range sensing is used to
 - A) Measure the robot's internal state
 - B) Determine distance using light reflection
 - C) Measure temperature differences
 - D) Estimate force applied by the robot
11. From which of the following is known as non-contact sensor
 - A) Strain guage
 - B) Limit switch
 - C) Capacitive proximity sensor
 - D) Tactile sensor
12. Which sensor operates by detecting eddy currents induced in metal objects?
 - A) Capacitive sensor
 - B) LVDT sensor
 - C) Inductive proximity sensor
 - D) Optical encoder
13. When selecting an electric actuator for a high-torque, low-speed application, which motor would you consider
 - A) SMA Actuator
 - B) DC servo motor
 - C) Linear motor
 - D) AC motor
14. What is the primary function of an H-bridge circuit in DC motor control?
 - A) To regulate voltage levels
 - B) To amplify the current for increased torque
 - C) To enable bidirectional control of the motor
 - D) To control the speed of the motor
15. What is the advantage of using brushless DC motors over brushed DC motors?
 - A) They have lower torque at high speeds
 - B) They have higher wear and tear
 - C) They are less efficient
 - D) They require less maintenance
16. Which type of actuator is most suitable for heavy-load applications?
 - A) Electric actuator
 - B) Pneumatic actuator
 - C) Hydraulic actuator
 - D) Piezoelectric actuator
17. Which robots are designed to provide companionship and emotional support?
 - A) Surgical Robots
 - B) Telepresence Robots
 - C) Social Robots
 - D) Rehabilitation Robots
18. What is a key attribute of bio-inspired robotics?
 - A) Replicating human speech
 - B) Using solar energy
 - C) Being inspired by nature
 - D) Ability to cook
19. Which of the following terms refers to robots specifically designed to work safely alongside humans in a shared workspace
 - A) Autonomous Robots
 - B) Collaborative Robots
 - C) Soft Robotics
 - D) Industrial Automation Robots
20. Fixed-wing UAVs are known for:
 - A) Hovering capabilities
 - B) Long flight times
 - C) Vertical landing
 - D) Simple design

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

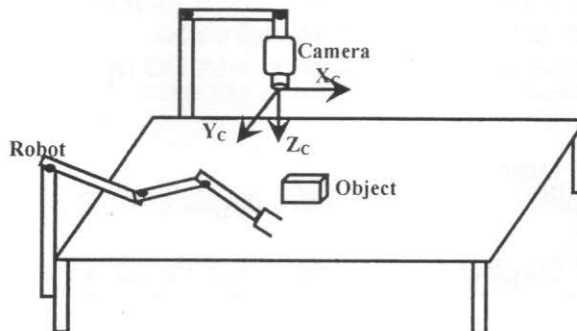
1. a) What are the typical features and applications of following Robot configurations?
 - i. Cartesian Coordinate robot
 - ii. Cylindrical Coordinate robot
 - iii. Spherical Coordinate robot
 - iv. Jointed arm robot
 - v. SCARA robot

8 L*2 1 1

- b) A robotic workstation has been setup with a TV camera as shown in Fig. The camera can see the robot coordinate system where base of a six joint robot is attached. It can also see the centre of an object (Cube) to be manipulated by the robot. If local coordinate system is established at the centre of the cube, this object as seen by the camera can be represented by a homogeneous transformation matrix ${}^{camera}T_{cube}$. If the origin of the base coordinate system as seen by the camera can also be expressed by the homogeneous transformation matrix ${}^{camera}T_{base}$.

$${}^{camera}T_{object} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^{camera}T_{robot} = \begin{bmatrix} -1 & 0 & 0 & -10 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & -1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine the position and orientation of cube coordinate system with respect to the base coordinate system.



8 L3 2 2

2. a) A cylindrical robot has a prismatic joint with a range of travel of 800mm. The control memory for this joint has 10-bit capacity. The arm's mechanical inaccuracy shows a standard deviation of 0.1 mm around the taught point. Determine the following,

- The control resolution.
- The accuracy.
- The repeatability.

8 L3 1 2

- b) Discuss the role of robot end-effectors and the different types of end-effectors used in various applications with neat sketch.

8 L2 2 1

3. a) What is the typical anatomy of a robotic manipulator? Explain with a neat sketch

8 L1 1 1

- b) For the gripper mechanism shown in Fig.Q3(b), obtain the actuating force F_a if the gripping force F_g is 20N.

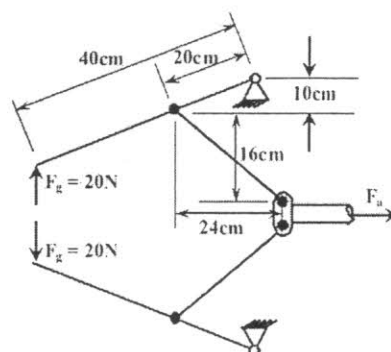


Fig. Q3(b)

8 L3 2 3

Unit – II

4. a) What are Proximity sensors? Explain the working principle of following sensors:
 i. Inductive proximity sensor
 ii. Capacitive proximity sensor
 b) Explain the working principles of hydraulic actuators. Discuss their advantages and disadvantages in industrial robotics applications.
5. a) Differentiate between following types of sensors and give examples for each type:
 i) Proprioceptive and Exteroceptive sensors
 ii) Active sensors and Passive sensors
 b) With the help of a H-Circuit, Explain the switching sequence for Full stepping bipolar permanent magnet stepper motor
6. a) Describe the working of an Absolute Rotary Encoder with the help of a neat sketch. Explain how Binary to Gray code conversion is performed.
 b) Sketch and explain working principle and control of brushless DC Motor with switching circuit.

8	L1	3	1
8	L1	4	1
8	L2	3	1
8	L2	4	2
8	L2	3	2
8	L2	4	2

Unit – III

7. a) What is Legged Locomotion? Explain. What are its advantages over wheeled Locomotion?
 b) What are the types of Medical or Surgical Robots? What are the advantages and limitations of Medical Robots?
8. a) Discuss the current trends in robotics and explain the various new applications where robots are being deployed beyond traditional industrial settings
 b) What should be the direction of the movement of propellers in a quadcopter for four primary movements? Explain with sketch.

8	L2	4	1
8	L2	4	1
8	L2	4	1
8	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
